Secrets

Securing Sensitive Data in K8s

Need

Various services in K8s will need sensitive information
→ *eg* credentials, authentication tokens

Storing them in configuration files in git often means too much access

K8s **Secrets** store sensitive information in the cluster
Values only accessible to resources in the same namespace

Creating a Secret

Secret resources are not tracked with YAML manifests

That would defeat the point

They are usually created via the command line:

```
kubectl create secret -n namespace generic my-secret-name \
  --from-literal=key1=value1 --from-literal=key2=value2
```

Secrets can contain multiple key/value pairs

Accessing Secrets

Secrets can be injected into containers as environmental variables:

```
spec:
  containers:
    - # ...
      env:
        - name: SECRET_TOKEN
          valueFrom:
            secretKeyRef:
              name: my-secret-name  # Name of secret
              key: key1             # Key within secret
```

Limitations

Secrets are stored unencrypted by default in *etcd*
⇒ Enable encryption at rest

All users and service accounts have access to Secrets' values
⇒ Use RBAC to limit access to Secrets

⇒ Consider using an external secrets vault along with the External Secrets Operator